

# How Embodied Conversational Agents Should Enter Your Space?

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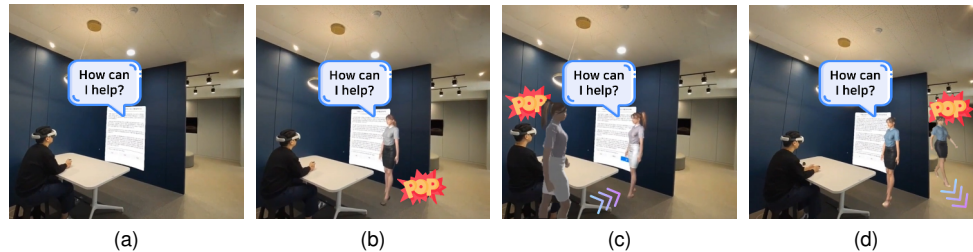


Figure 1: We compared four different entrance styles of embodied agents. The experimental conditions are categorized based on their plausibility as actions that a real person could perform: (a) baseline voice-only agent, (b) agent appearing suddenly in the air within the user's view (low plausibility), (c) agent emerging from behind a space where it is ambiguous whether a person could actually come out (moderate plausibility), and (d) agent walking in from an open area (high plausibility).

## ABSTRACT

Embodied conversational agents (ECAs) capable of nonverbal behaviors have been developed to address the limitations of voice-only assistants. Research has explored their use in augmented reality (AR), suggesting they may soon interact with us more naturally in physical spaces. However, the question of how they should enter the user's space when summoned remains under-explored. In this paper, we focused on the plausibility of ECAs' entering action into the user's field of view in AR. We analyzed its impact on users' perceived social presence and functionality of the agent. Our results indicated that the plausibility of the action significantly affected social presence and had a marginal effect on perceived functionality.

**Index Terms:** Augmented Reality, Embodied Conversational Agent, Behavioural Plausibility, User Perception, Social Presence.

## 1 INTRODUCTION

With the advent of smartphones equipped with voice assistants like Siri and Bixby, users can perform simple tasks through conversational interactions with the assistant. However, voice-only interactions often feel less natural than human communication due to the absence of non-verbal cues. Non-verbal cues, such as eye contact, gestures, and body language, are integral to human communication, facilitating rapport and trust during conversations [1]. To mitigate these limitations, researchers have developed embodied conversational agents (ECAs) that incorporate non-verbal behaviors, including gestures and facial expressions [2]. Recent studies have extended these interactions beyond 2D displays to immersive environments, such as virtual reality and augmented reality (AR) [3, 4, 5].

The development of head-mounted displays (HMDs) enables users to interact with familiar voice assistants in immersive environments, allowing these assistants to be manifested in embodied forms. This transformation enables users to converse and interact with them more naturally, similar to interactions with real humans. However, if an ECA remains visible when not in use, it may ob-

struct the user's view, so it should only appear when necessary. To our knowledge, no research has explored how ECAs, initially absent in AR, should be introduced into the user's field of view.

This study aims to investigate methods for ECAs to appear in users' fields of view when summoned for tasks in AR, especially in confined spaces, such as office rooms, where the ECA's entrance behavior might be constrained by spatial limitations. Previous research suggests that violations of physical laws, such as ECAs moving through walls, diminish users' perceptions of social/co-presence [3], and that the perceived functionality of ECAs can be influenced by their behavior [4]. Consequently, this study examines how the plausibility of ECA entrance styles influences users' perceptions of social presence and the functionality of the ECA.

## 2 METHODOLOGY

### 2.1 Design

We employed a within-subjects design with four conditions: three ECA entrance styles and a voice-only baseline condition. The order of the conditions was randomized, and 3D human models were also randomly assigned to conditions involving embodied agents. Participants engaged in a problem-solving task based on the provided paragraphs. In one condition, three paragraphs were presented with three questions, one of which could not be directly inferred from the text. Participants had to summon the agent by calling it their preferred name when encountering these questions. Depending on the condition, the agent responded either with voice only or with both voice and embodiment. This design ensured that each participant summoned the agent at least three times per condition.

We used the Meta Quest 3 with Full-Color Passthrough enabled to provide a video see-through AR experience. A virtual question board and ECA were augmented into the real space (See Fig. 1).

### 2.2 Conditions

We prepared three different entrance styles for the embodied agents, along with a baseline voice-only condition, as follows:

**Baseline (B):** This condition replicates an existing voice-only assistant. When summoned, the agent responds verbally and deactivates after answering without any visual indications (See Fig. 1a).

**Low-Plausibility (LP):** The ECA appears in the air next to the question board within the user's view and also disappears from the same location. This scenario represents low plausibility as a physical entity, and the user could notice this odd moment (See Fig. 1b).

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**Moderate-Plausibility (MP):** The ECA emerges from behind the user and walks to the designated location to interact with the user. Once the ECA answers the user's question, it returns to the emergence position and disappears. Note that users are aware that the space behind them is confined and vacant, making this scenario implausible if the ECA were a physical entity. However, users are less likely to observe this odd moment (See Fig. 1c).

**High-Plausibility (HP):** The ECA emerges from the hallway behind the wall connected to the external space, enters the room through the passage leading to it, and stands at the designated location. This condition represents the most plausible entrance style as a physical entity. After answering, the ECA exits through the hallway and disappears. The key difference from the MP condition is that the ECA's activation and deactivation occurred behind the wall, where the user's field of view is completely blocked (See Fig. 1d).

### 2.3 Participants

This study was approved by the Institutional Review Board (IRB) of the authors' institution (No. 2024.23.HR). We recruited 40 participants from a local university. Two participants did not complete the experiment, leaving data from 38 participants (21 males, 17 females; M age = 23.66 (SD = 1.98)) for analysis. Participants reported experience with ECA (M = 1.89 (SD = 1.19)) and voice assistants (M = 3.95 (SD = 1.88)).

### 3 RESULTS

We used five items to assess social presence [6], and three items for functionality [7]. To compare the effects of entrance style, we used mean ratings across participants (Cronbach's  $\alpha = 0.757$  and  $0.906$ , respectively). We performed a Friedman test to assess the main effects of entrance style, followed by Wilcoxon signed-rank tests for pairwise comparisons with Bonferroni correction. The significance level for all statistical analyses was set at  $0.05$ . The results indicated a statistically significant main effect of entrance style on social presence ( $\chi^2 = 44.652$ ,  $p < 0.001$ ). For functionality, the main effect was marginally significant ( $\chi^2 = 7.563$ ,  $p = 0.056$ ).

The results indicated that the embodiment could impact social presence and functionality (See Fig. 2). Within the embodied agent conditions, the participants rated the social presence of the LP condition significantly lower than both MP and HP. The absence of locomotion and noticeable physical inconsistencies—or what could be described as physically implausible behavior—likely contributed to this outcome. This finding is consistent with previous studies, which reported that ECAs with locomotion were rated higher in social presence compared to those without [4]. Additionally, ECAs that violated physical rules or seemed physically unrealistic were associated with reduced social presence [3, 5].

Although the Friedman test did not indicate a statistically significant main effect for functionality, we conducted Wilcoxon signed-rank tests to explore potential pairwise differences among conditions. For functionality, there were no significant differences within the embodied agent conditions. However, all the embodied agent conditions received significantly higher scores than the voice-only agent (See Fig. 2b). This suggests that even when an agent provides the same functions to complete the tasks, the presence of a visible body and actions, including entrance styles, can influence how users perceive its capabilities.

### 4 CONCLUSION

In this paper, we explored how the plausibility of entrance styles affects users' perceptions of social presence and the functionality of ECAs in AR. We compared three ECA entrance styles with a voice-only baseline condition. Our findings revealed that high-plausibility entrance styles significantly enhance social presence. However, as the HP condition requires systems capable of environmental recognition and collision-free path generation, the MP condition, with

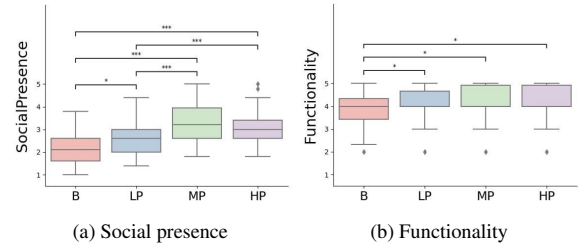


Figure 2: Results of the measurements: (a) Social presence and (b) Functionality (\*:  $p < 0.05$ , \*\*:  $p < 0.01$ , \*\*\*:  $p < 0.001$ ).

fewer spatial constraints, offers a practical alternative, particularly for confined spaces. These results provide valuable insights for implementing ECAs in HMDs, such as Apple's Vision Pro, and present a paradigm for designing ECAs where entrance styles can be tailored to specific applications or use cases.

However, certain aspects require further investigation. First, we focused on a task with significant visual load, such as reading and selecting answers, which may have reduced attention to the agents, limiting generalizability. Future research should explore tasks with less visual load, allowing more focus on the agents. Additionally, we employed female agents, which could have introduced gender bias. The use of a human-like agent might have affected perceptions of behavioral plausibility. With non-human agents, users' evaluations of entrance style plausibility and consequently their perceptions of social presence and functionality might differ. Future studies should explore how varying anthropomorphism, gender, and other aspects of agent appearance influence user perceptions.

### ACKNOWLEDGMENTS

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